Examination of Target Classification for Millimeter-Wave Radar

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0.Abstract

As an alternative to camera-based object identification, which performs poorly in adverse weather, object classification was attempted using millimeter-wave radar and machine learning. Despite lower angular resolution, the radar signals enabled identification of pedestrians, bicycles, and vehicles with 90% accuracy under ideal conditions, confirming the effectiveness of radar signals.

1.introduction

Pedestrian detection using sensors installed on vehicles or roadside infrastructure has been widely studied. However, image-based methods suffer from degraded performance in adverse weather such as heavy fog. Radar, which is more robust to weather and can directly measure distance, provides less angular resolution compared to cameras, resulting in limited information. This limitation can be addressed by temporally accumulating radar-based features, enabling reliable pedestrian detection even in poor weather. We examined an object classification method that accumulates radar data and uses machine learning to distinguish pedestrians, bicycles, and vehicles. This paper presents the classification approach and evaluation results under ideal conditions, demonstrating the potential of radar-based object identification.

2. mm-wave radar

2.1 Principles of Radar

Radar can measure the distance, angle, and speed of a target. Distance is calculated from the round-trip time of the transmitted radio wave. For a target 30 meters away, the round-trip time is only about 0.5 microseconds, which is too short to measure directly. Instead, the FMCW (Frequency-Modulated Continuous Wave) method is used, where the transmission frequency increases linearly over time. The frequency difference between the transmitted and reflected signals, caused by the time delay, allows calculation of the distance. Angle estimation involves changing the direction of transmission and reception. This can be done mechanically by rotating the antenna, or electrically using a phased array antenna. The latter enables MIMO radar systems, which require fewer antenna elements. Speed is determined by detecting phase changes in the received signal. The phase depends on the signal wavelength and the distance to the target, changing 360 degrees for every half-wavelength of distance change. As the target moves, this phase varies over time, and its rate of change reveals the target's speed.

2.2 Millimeter wave radar method

Automotive radar often uses unlicensed millimeter-wave systems classified as specified low-power radio stations. The short wavelength enables compact and lightweight antennas, and the FMCW (Frequency-Modulated Continuous Wave) method, which places minimal load on the transmitter, is widely used for distance measurement. The radar used in this study also follows this approach. For angle measurement, MIMO (Multiple-Input Multiple-Output) systems are adopted, leveraging compact antennas and eliminating the need for mechanical rotation.

2.3 Information that can be obtained

The radar in this study acquires velocity in addition to range and angle. Specifically, it extracts a reflection intensity distribution in a 3D space (defined by range, angle, and velocity axes) from the received signal, as shown in Figure 1 (left). From this distribution, target point clouds are detected by thresholding, assuming that useful targets exhibit high reflection intensity. The detection threshold is calculated for each point in the 3D space based on the surrounding reflection intensities. The coordinates (range, angle, velocity) of the detected points constitute the target information. While the radar extracts target positions in terms of range and angle, our proposed target identification method utilizes Cartesian coordinates because they allow for more intuitive handling of information crucial for identification, such as depth and width. Therefore, a transformation to the Cartesian coordinate system is performed as a preparatory step for identification.

3. Processing by object label

3.1 Machine learning for object labeling

In the machine learning-based identification process, as shown in Figure 2, target features extracted from radar data by preprocessing (Section 3.2) are fed into an identification rule, which outputs an identification label (pedestrian, bicycle, or vehicle). This identification rule is generated by a learning algorithm, such as Support Vector Machine (SVM) [2], from training data. The training data consists of pairs of features and their corresponding ground truth labels; these features are extracted from pre-recorded radar data for each target type (pedestrian, bicycle, and vehicle) using the same preprocessing method. Generally, a larger amount of training data allows for the construction of a model capable of handling diverse situations.

3.2 Pretreatment

Preprocessing extracts features useful for object classification from radar-detected targets, as illustrated in Fig. 3. After thresholding, multiple reflection points are often detected for a single object. We group these points into clusters based on their spatial proximity in 3D space. This clustering enables the extraction of primary features for each object at every sampling time. In addition, features that span multiple time samples—such as temporal changes—are also effective for classification. Therefore, we track each object over time by associating clusters across sampling intervals. The extracted primary features are accumulated over time, and their statistical values are used as secondary features for input to the machine learning model.

As shown in Fig. 4, primary features represent the geometric characteristics of object clusters. The figure compares 3D point clusters of a pedestrian and a vehicle at a specific time. Their spatial distributions differ: for example, the vehicle cluster tends to extend further in the depth direction compared to the pedestrian. Typical speeds also vary significantly between the two. While individual cases differ depending on the situation, such features are considered statistically important for classification.

4. evaluation

To investigate the potential of object classification using millimeter-wave radar, we collected separate datasets for vehicles, bicycles, and pedestrians under ideal conditions without obstacles. First, we verified whether effective features for classification could be extracted from the data. Then, we trained a classification model using those features and evaluated its accuracy.

4.1 Data Collection Environment

As shown in Fig. 5, data were collected in an open area free of buildings and obstacles. The movement directions of the targets included approaching and receding from the radar, as well as crossing from right to left and from left to right. For each direction, multiple scenarios were recorded by varying walking speed and movement conditions.

4.2 Features Used for Evaluation

To verify whether meaningful features can be extracted from the radar data, we arranged the primary features in time series and analyzed their temporal variations. Figure 6 shows the time evolution of the depth-wise distribution of reflection points for a slowly receding vehicle and pedestrian. The lines in the figure represent the maximum, average, and minimum depth positions of reflection points within each cluster. The difference between the maximum and minimum depth values indicates that the vehicle has a wider spread of reflection points than the pedestrian. Additionally, the difference in depth spread between a bicycle and a pedestrian remains relatively stable over time. Figure 7 shows the temporal change in the velocity distribution of reflection points for a vehicle and a pedestrian. The lines in the figure represent the maximum, mean, and minimum velocities within each cluster. At any given time, the spread of the vehicle's velocity distribution is narrower than that of the pedestrian's. Furthermore, the vehicle's velocity remains stable, whereas the pedestrian's velocity fluctuates over time.

4.3 Conditions

The accumulation time for primary features was set to 0.5 seconds. The features for our model were the temporal variations in the "spread" and "mean" of reflection point clusters along the range, velocity, and horizontal axes. Although mean velocity is effective for classification, it was excluded from the training data to avoid the risk of the model overfitting to velocity, given the limited number of scenes in our dataset. As shown in Figure 8, radar data from each scene was divided into 0.5-second blocks. Primary features within each block were temporally accumulated to calculate the final features. For the dataset, we extracted a total of 395 blocks: 127 from vehicles, 99 from bicycles, and 169 from pedestrians, all moving individually. These features are classified into one of the three classes (vehicle, bicycle, or pedestrian) based on pre-trained identification rules. These rules were generated using training data composed of feature-label pairs. The classification accuracy is defined as the ratio of correct classifications to the total number of classifications performed every 0.5 seconds. For instance, in the example shown in Figure 8, the accuracy is 80%, with 8 correct classifications out of 10. This indicates that the temporal variations in primary features contain effective information for distinguishing among vehicles, bicycles, and pedestrians.

4.4 Results

The classification results are shown in Table 1. Due to the limited number of data blocks, we used leave-one-out cross-validation (3) for evaluation. The key findings are as follows: Vehicles and Pedestrians: Both were correctly classified with high accuracy (>90%), and misclassifications between these two classes were below 3%. Bicycles: The correct classification rate was below 90%, with over 10% being misclassified as pedestrians. This classification accuracy is not necessarily superior to that of image-based object identification methods. Future work will focus on two areas: Improving Classification Accuracy: This will involve retraining the model with more diverse data, including scenarios with adverse weather, occlusions, and objects with complex movements. We will also explore methods that accumulate information over multiple blocks for decision-making. Exploring Applications: We will investigate specific applications that can leverage the inherent advantages of radar, such as its weather resistance and direct velocity detection capabilities.

5. Conclusion

Using machine learning with features extracted from millimeter-wave radar data, we achieved approximately 90% accuracy in classifying pedestrians, bicycles, and vehicles under low-noise conditions. Features of cars and pedestrians were distinctly different, while bicycles exhibited characteristics of both, indicating that radar data contains useful information for object classification.

However, accuracy may degrade in complex scenes involving occlusion or parallel movement, such as bicycles riding alongside pedestrians or pedestrians behind guardrails. To address this, we are collecting data under various conditions and plan to improve accuracy using more advanced learning algorithms. This technology is also applicable beyond traffic scenarios. In environments where privacy concerns prohibit the use of cameras, millimeter-wave radar is a suitable alternative. Such settings present promising opportunities for pedestrian detection using radar-based classification.

ChatGPT 4-o と Google AI Studio Gemini 2.5 Pro Preview 05-08 を利用してそれぞれに「この文章から 1、重複表現を減らす。 2、無駄な言葉や言い回しをなくす。 その上で英訳してください。 すべての過程を表示してください」と入力した。出力されたものをさらに原文と英文それぞれを Gemini Pro 2.5 に入れた上で、「この和訳を以下に 2 つ示します。どちらの翻訳がよいかそれぞれ元の文章を参照したうえで検討してください。」と入力し、よいと判断した方を適用した。